Jasser Chaabane

Robotics Engineering Intern

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Professional Profile

I'm a final-year Mechatronics engineering student who believes robots can make a difference for the people who rely on it like for example in elderly healthcare. I like to go beyond my comfort zone: I taught myself robotics software from scratch and I enjoy turning new ideas into things that really work. I'm open to fresh ideas, happy to challenge the usual way of doing things, and just as comfortable working independently as in a **multicultural, interdisciplinary team**. What I want most is to help solve **real problems in Robotics technology**, from the first idea to the final result. I'm looking for a **final-year internship** and I'm **willing to relocate**.

Education

National Engineering School of Sousse (ENISO) — **2024 – 2027 (expected)**

*National Engineering Degree (Master's level), Mechatronics: robotics,**CNC machining**, embedded systems, study program* — *Sousse, Tunisia*

Preparatory Institute for Engineering Studies of Monastir (IPEIM) — **2022 – 2024**

Intensive Preparatory Classes for Engineering Schools (Mathematics & Physics) — *Monastir, Tunisia*

Academic Projects

Autonomous Mobile Robot (AMR) — ROS2, SLAM & Nav2 Navigation, Self-Taught — **2025**

ROS2 · Gazebo · RViz · SLAM Toolbox · Nav2 · TF2 · C++ · Python · LiDAR · Linux

- Designed and simulated a differential-drive robot under **ROS2** in **Gazebo**: real-time LiDAR SLAM mapping and dynamic path planning with **Nav2**, <5 cm accuracy in simulation.
- Learned ROS2, SLAM and Nav2 on my own (not in the ENISO curriculum): nodes, topics and **TF2** frame trees.

Vision-Guided Robotic Pick-and-Place — Grasping Randomly Oriented Parts — **2025**

Webots · Computer Vision · Cameras · 3D Pose Estimation · Inverse Kinematics · MATLAB · SolidWorks

- Built a full **perception-to-grasp pipeline** in simulation: camera-based detection of parts on a moving conveyor, **3D pose estimation** of each part, and **inverse kinematics** to grasp and sort them.

DC Motor Speed Control — STM32 & Discrete Power Stage — **2024**

STM32 · Embedded C · Hardware PWM · IR2103 · IRFZ44N

- Closed the loop from microcontroller to actuator: hardware PWM from STM32 timers driving a discrete half-bridge (IR2103 + IRFZ44N), solid **embedded systems** foundation for robot actuation.

Professional Experience

Nexor Robotics — **Aug 2026**

Robotics Engineering Intern — LLM-Based Coordination of Autonomous Mobile Robots (AGV) — *Sahloul, Tunisia*

- Designed an industrial **AGV** from scratch for a **ROS2** + **Gazebo** simulation: differential-drive chassis modeled in CAD and described in **URDF/Xacro**, with 2D **LiDAR**, camera and odometry, navigated with **Nav2**.
- Built a coordination system where an operator types a mission in natural language in a **chatbot** and an **LLM** splits it into tasks and assigns them to a fleet of 2 to 6 AGVs, based on each robot's position and battery level.
- Added a **plan verifier** that checks every LLM plan (strict **JSON** schema, zone conflicts, low battery, unreachable zones) before it reaches the robots, so the model never drives the AGVs directly.
- Defined the JSON messages exchanged between the chatbot, the LLM and the ROS2 nodes, and reassigned tasks to the other robots when an AGV ran low on battery or failed.

TAV Tunisia — Monastir Habib Bourguiba International Airport — **Jul 2026**

Engineering Intern — Real-Time Multi-Zone IoT Monitoring — *Monastir, Tunisia*

- Designed and deployed a real-time monitoring system of 25 **sensor** channels (CO_2, temperature, humidity, light, occupancy) on MQTT (Mosquitto), with end-to-end latency below 200 ms.

LEONI Wiring Systems — **Jul – Aug 2025**

Immersion Internship — Industrial Maintenance & Quality Assurance — *Sousse, Tunisia*

- Electrical quality control on automated test benches for automotive wiring harnesses (IATF 16949 / ISO 9001), in a strict, quality-driven industrial environment.

Technical Skills

- **CAD & Rapid Prototyping:** SolidWorks (CSWP) 3D modeling and parametric assemblies; **Autodesk Fusion**,rapid prototyping of electronics and actuation; testing and implementation of solutions,**3D printing technologies**.
- **Robotics & Automation: ROS2**, **Gazebo**, RViz, **Nav2**, SLAM Toolbox, TF2, Webots, Inverse Kinematics, Path Planning, Odometry, multi-robot coordination, simulation of dynamic environments, automated test benches.
- **Sensors & Cameras:** LiDAR, camera-based detection, 3D Pose Estimation (simulation), multi-sensor IoT acquisition (CO_2, temperature, humidity, light, occupancy) over MQTT.
- **Computer Vision:** , OpenCV, object detection, 3D Pose Estimation; NVIDIA DLI AI Anomaly Detection.
- **Embedded Systems:** STM32, ESP32, FreeRTOS, Embedded C, hardware PWM motor control, CAN, I2C, SPI, UART,electronics assembly, soldering, and wiring.
- **Programming Languages: C++**, **C**, **Python**,**MicroPython**, MATLAB.
- **Machine Learning:** scikit-learn,NumPy,Pandas,Matplotlib, Supervised/Unsupervised Learning

Certifications

- **SolidWorks Certified Professional (CSWP):** Advanced 3D modeling and parametric assemblies (Dassault Systèmes).
- **Specialized Certifications:** Python 3 for Robotics (The Construct) • NVIDIA DLI: Applications of AI for Anomaly Detection • FreeRTOS & Embedded C (CSF).
- **Machine Learning with NumPy, pandas, scikit-learn, and More** (Educative, 15 hours, Oct 2026).
- **GCI World, University of Tokyo** *(3-month program, in progress, Sep–Dec 2026):* data competition and capstone project.

Activities & Languages

- **2nd Place, ECPP 2.0 National Hackathon:** STM32F407 firmware in 48 h.
- **1st Place, Deathring Combat Robotics Challenge** (ENISO).
- **1st Place, IoT Smart Challenge Hackathon** (Club Orange).
- **Clubs:** ENISO Companies Forum organizing committee • Aerotronix, ACM, Eureka • Rotaract Tunisia.
- **Languages:** English (fluent / C1 — **TOEIC: 960/990**) • Arabic (native) • French (bilingual / C1) • German (basic / A1).